

Bristlecone: an F# library for the long-term ecological detective

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Summary

Long-term ecology (LTE) seeks to extend our understanding of ecological systems to decades to centuries, beyond the observational period. LTE researchers employ environmental proxies (e.g., microfossils, wood rings) to reconstruct variability in biodiversity and environments over decades to centuries. Disparate dating methods (e.g., radiocarbon dating) and representations of time are thus usually required for interpretation. Although the long view is integral to comprehensively understand ecosystem responses to environmental change, there has historically been limited application of causal inference within LTE (Katherine J. Willis et al., 2007).

Mechanistic modelling approaches may be used to infer the form and strength of ecological processes. In *The Ecological Detective*, Hilborn & Mangel (1997) present a toolkit for confronting ecological models with data. Here, I developed a modelling framework to enable LTE researchers to apply the ecological detective approach to proxy time-series for causal inference. Key capabilities include: a human-readable Domain Specific Language (DSL) for model definition with enforced dimensional correctness; model composition and scaffolding competing hypotheses; and confrontation of models with time-series of disparate dating methods.

Statement of need

Research purpose

Bristlecone's purpose is to enable researchers to build causal knowledge of the role of ecological mechanisms over decades to centuries, through confrontation of models with proxy data. LTE is essential in predicting biodiversity responses to climate change, as we may (indirectly) observe (a) ecological processes that operate over longer timescales (e.g., soil development (McLauchlan et al., 2017)); (b) biodiversity responses to broader environmental envelopes; and (c) ecological stable states and resilience (K. J. Willis et al., 2010). Although LTE research has often been descriptive, causal understandings are essential for establishing predictive capability. Causal understanding may be achieved through combining multiple lines of evidence, including (a) experimentation, (b) identifying spatial-temporal associations in observational data, and (c) mechanistic investigation. Without experimental approaches in LTE, surrogates such as space-for-time substitution can misrepresent the rate and order of processes (Elmendorf et al., 2015); thus, mechanistic investigation is essential.

In contemporary ecology ('neo-ecology'), the 'ecological detective' is an approach for causal inference (Hilborn & Mangel, 1997). The approach requires: (1) examining theory to understand plausible mechanisms; (2) understanding process and observation uncertainties; and (3) confronting plausible alternatives with data. Taking causal mechanisms as processes "*along which a signal can be propagated to produce a response*" (Grace et al. (2025), p5), the ecological detective approach may be applied to infer causal mechanisms using LTE time-series.

41 *Bristlecone* is a toolkit that enables LTE researchers to utilise the ecological detective approach
42 through: (1) a human-readable, succinct declarative grammar for the ‘ecological detective’
43 workflow; and (2) implementations of key components required to use the grammar. The
44 toolkit includes a grammar of time that reflects the dating methods used within LTE and
45 neo-ecology; it thus targets neo-ecologists who would like to integrate neo-ecological and LTE
46 time-series for multi-scale analysis.

47 *Bristlecone* occupies a distinct niche: although tools exist for data wrangling and inference,
48 *Bristlecone* is a higher-level grammatical layer focused on correctness in model construction
49 from an ecological perspective. General-purpose inference engines (e.g., Stan, JAGS) lack
50 dimensional correctness and often require manual construction of temporal concepts. Stan’s
51 language, for example, does not support dimensional correctness (Stan Development Team,
52 2025). *Bristlecone* provides that missing layer within a formalised ecological detective workflow.

53 State of the field

54 The ecological community predominantly utilises R. Within LTE, key statistical approaches
55 include transfer functions, generalised additive models, and canonical correspondence analyses.
56 Proxy-specific mechanistic approaches have also been applied; the landscape models REVEALS
57 (Sugita, 2007a) and LOVE (Sugita, 2007b) simulate taxon-specific biomass from pollen fluxes.
58 However, mechanistic inference from LTE time-series has been limited (e.g., Jeffers et al.
59 (2011)). In dendrochronology, (semi-)mechanistic approaches are more prevalent, such as
60 inferring temperature/moisture limits to tree growth using VS-Lite (Tolwinski-Ward et al.,
61 2011).

62 *Bristlecone*’s two unique contributions are: (1) integrating the ecological detective workflow
63 within a single conceptual framework; and (2) providing ecological clarity through dimensional
64 correctness. *Bristlecone* utilises F#’s type system and expressiveness to enforce dimensionally-
65 consistent ecological models and promote human readability, improving transparency and
66 reproducibility of supplementary material. As the core design relies on F# language features
67 that could not be replicated within R, *Bristlecone* was designed from scratch. However, a
68 compositional approach was sought whereby components may be substituted where beneficial
69 alternatives exist (e.g. optimisers). The F# R type provider (FsLab, 2012) presents an avenue
70 for embedding R libraries and graphics via typed R access.

71 *Bristlecone*’s target audience is long-term- and neo-ecologists who wish to explore mechanistic
72 modelling. The library aims to reduce computational knowledge required to utilise the ecological
73 detective approach within LTE research.

74 Software design

75 Although F# is uncommon in ecological research, I considered that an F#-based domain-
76 specific language could be intuitive for ecologists with R experience. I sought to include
77 the seven stages of the ‘ecological detective’ workflow: (1) definition of time-series models;
78 (2) specifying model-fitting methods (“estimation enigmes”); (3) composition of alternative
79 model hypotheses; (4) identifiability testing; (5) model-fitting; (6) model-selection; and (7)
80 diagnostics.

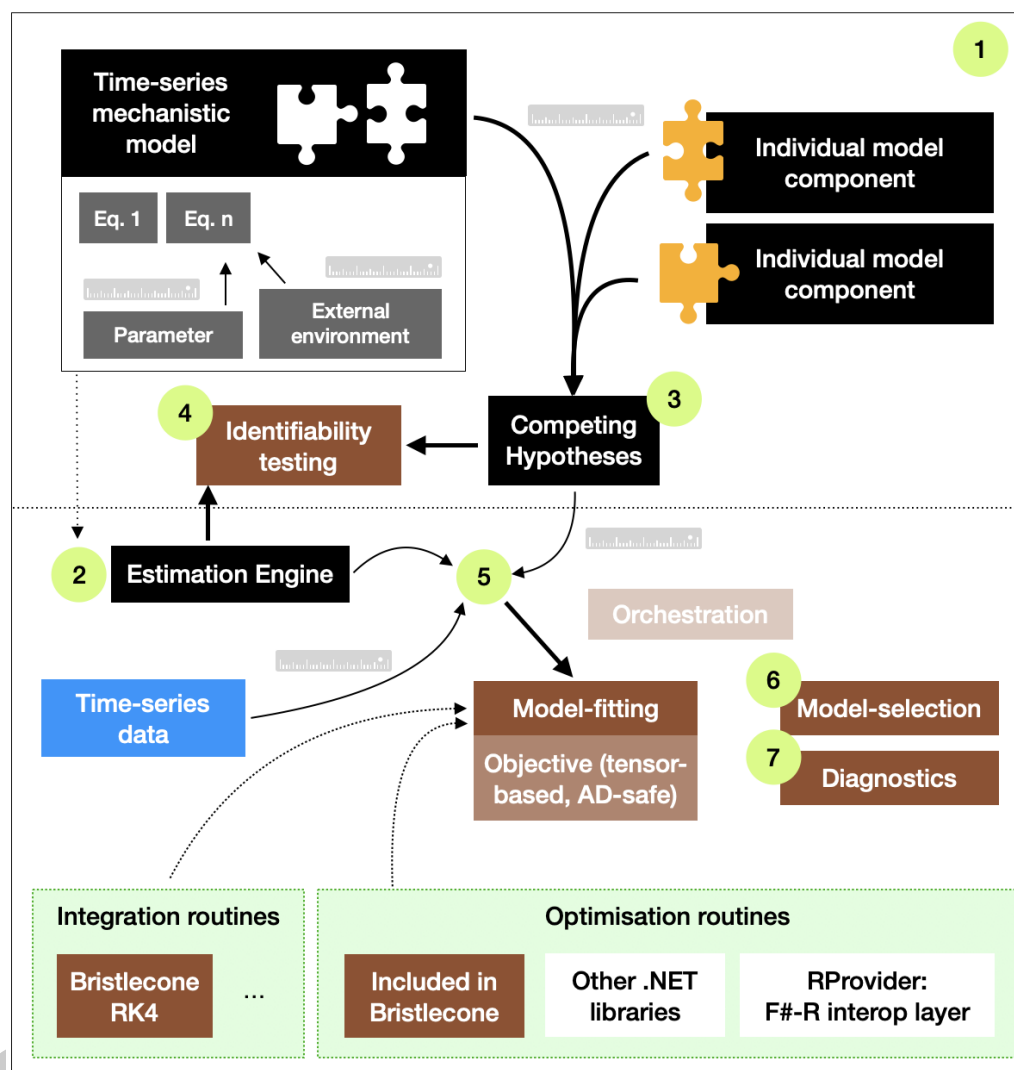


Figure 1 Conceptual framework. Green circles indicate stage of analysis. Box colours: brown = Bristlecone core functions; black = user-defined elements using the Domain Specific Language; blue = user-supplied data; white = external functions.

A core requirement was that ecological models should utilise F# type safety and units of measure such that all models, their composition, and methods are dimensionally consistent. I designed a DSL that ensures dimensional consistency across equations, parameters, external forcings (e.g., climate), and methodological approaches, supporting the scientific process by raising errors during compile-time prototyping.

To demonstrate, fish stocks (see: *The Ecological Detective*, Chapter 10) may be represented in discrete-time by:

```
module Schaefer =

    let B = state<kton> "biomass"
    let I = measure<1> "CPUE" // Catch per unit effort
    let C = environment<kton> "catch"

    let r = parameter "r" Positive 0.01</year> 1.0</year> // Intrinsic growth rate
    let K = parameter "K" Positive 100.<kton> 2000.<kton> // Carrying capacity
    let q = parameter "q" Positive 1e-6<1/kton> 1e-2<1/kton> // Catchability
```

```

let B0 = P K

let dt = Constant 1.<year>
let ``B_est[t+1]`` =
  let n = State B + P r * State B * (Constant 1. - State B / P K) * dt - Environme
    Conditional (n .> Constant 0.<kton>) n (Constant 0.<kton>)

let It = P q * State B

let sigma = parameter "sigma_v" NoConstraints 0.01 1.0
let NLL = ModelLibrary.NegLogLikelihood.LogNormal (Require.measure I) sigma

let model =
  Model.discrete<year>
  |> Model.addDiscreteEquation B ``B_est[t+1]``
  |> Model.addMeasure I It
  |> Model.initialiseHiddenStateWith B B0
  |> Model.estimateParameter r
  |> Model.estimateParameter K
  |> Model.estimateParameter q
  |> Model.useLikelihoodFunction NLL
  |> Model.compile

```

92 Here, 'catch per unit effort' must be dimensionless as required by lognormal observation
 93 error, and `B_est[t+1]` must result in `kton`. If the model was continuous-time, `B` would equal
 94 `kton/year`. The entire model pipeline is unit-aware, preventing many model definition errors
 95 at compile-time. To compose alternative mechanisms, I designed a pipe-based workflow that
 96 turns nested model systems into a set of hypotheses while ensuring dimensional consistency
 97 between sub-components:

```

let hypotheses =
  ``base model``
  |> Hypotheses.createFromModel
  |> Hypotheses.apply ``geometric constraint``
  |> Hypotheses.apply ``plant-soil feedback``
  |> Hypotheses.apply ``temperature limitation to growth``
  |> Hypotheses.apply ``N-limitation to growth``
  |> Hypotheses.compile

```

98 I designed Bristlecone's *estimation engine* so that models and data may be combined across
 99 time concepts, with engines ensuring temporal consistency at compile-time. Radiocarbon,
 100 contemporary calendar, and AD/BC time-series may all be applied or combined with models of
 101 various temporal resolutions; the only requirement is that appropriate conversions are configured
 102 within the engine. The strength and importance of mechanisms over different timescales (e.g.,
 103 observational- versus centennial-time) may thus be investigated by combining ecological models
 104 with different (proxy) data sources.

105 Identifiability testing (assessing if the engine and model can infer known parameters) is essential
 106 for valid inference. I designed a unit-safe pipeline-based testing suite where synthetic time-
 107 series can be generated and applied to test identifiability of model-engine combinations. For
 108 model-fitting, I included orchestration for multi-threaded fitting. For model-selection I included
 109 Akaike weights, as *Bristlecone* currently focuses on information theoretic approaches. Finally,
 110 goodness-of-fit (root mean square error and n -step predictions) and other diagnostics were
 111 included. Resultant parameter estimates are unit-typed so unit consistency is carried to further
 112 analysis outside *Bristlecone*.

113 Integration with other libraries

114 I designed *Bristlecone* as a high-level workflow; it is thus extensible such that underlying
115 components, such as integration and optimisation routines, may be substituted. I chose to
116 enable gradient-based approaches by integrating automatic differentiation using DiffSharp
117 tensors (Baydin et al., 2015).

118 Three classes of optimisation routines broadly suited to ecological time-series were included,
119 while recognising that more appropriate alternatives for specific problems may be substituted.
120 First, amoeba-based methods have been applied to palaeoecological inference (Jeffers et al.,
121 2011) but are best suited to simpler likelihood surfaces. Second, I included Classical and Fast
122 Simulated Annealing [Lee (2015)](Szu & Hartley, 1987). Fast Simulated Annealing is more
123 performant at identifying global minima; *Bristlecone*'s implementation was previously applied in
124 dendroecology (Martin et al., 2021). Third, I included Monte Carlo methods including Filzbach,
125 a meta-heuristic specific to high-dimensional ecological models (Purves, 2016). As the existing
126 Filzbach library is unmaintained, and F# math libraries lack annealing, I implemented both
127 within *Bristlecone*.

128 Research impact statement

129 *Bristlecone* has been developed openly since 2018, maturing through research requirements. It
130 has been formatted for community use and contribution, including documentation, examples,
131 a benchmark suite, and contributor guidelines. Previous research applied multi-proxy
132 palaeoecological data to infer the role of soil nutrients in plant productivity through the
133 Holocene (Jeffers et al., 2011) using custom C; I reproduced that analysis as a *Bristlecone*
134 example.

135 *Bristlecone* was integral to Martin (2019) and Martin et al. (2021), where it was applied
136 to determine the role of soil nutrients in controlling Arctic shrub growth (*Bristlecone* v1).
137 Alternative hypotheses of nutrient limitation were confronted with wood ring and nitrogen
138 isotope time-series. That analysis is included as an example, formatted for *Bristlecone* v3.
139 *Bristlecone* is currently being applied to a pan-Arctic synthesis of Arctic palaeoecological
140 records (Martin et al., 2024).

141 AI usage disclosure

142 AI has not been used to write this manuscript.

143 No AI tools were used prior to August 2025. From then, the Microsoft Copilot macOS app was
144 used as an external tool to the development environment to suggest missing tests, refactoring
145 to code snippets, and potential approaches to proposed v3.0/3.1 features. Code suggestions
146 were critically examined and if beneficial implemented manually with human oversight. In
147 some cases, AI code suggestions were pasted into the development environment as a starting
148 point and then manually adapted. No agents have been used.

149 Acknowledgements

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